
Daniil Emtsev

AI Research Engineer | Zurich, Switzerland

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[GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

AI research engineer in Zurich. Real-time computer vision and language models for surgical training at VirtaMed, and generative AI that runs entirely in the browser. ETH Zurich MSc, a WACV paper and a patent application.

EXPERIENCE

Machine Learning Research Engineer

2022 - now

VirtaMed | Zurich

- Real-time computer vision in PyTorch for a browser app that analyses live camera input, with inference running fully client-side.
- LLM-based guidance and automated report generation integrated into surgical simulators.
- Transformer-based detection of anatomy in ultrasound, from 60% to 90% accuracy.
- A texture-enhancement system for visual realism, and a scalable pipeline that turns 3D simulation output into training data.
- End-to-end MLOps on Azure ML and DVC for reproducible training, versioning and deployment.
- Teaching: AI lectures at VirtaMed, supervising students from Swiss universities, and contributing to Innosuisse surgical proficiency projects.

R&D Software Engineer, Internship

Summer 2019

Data Analytics Group | Moscow

- Co-developed a scalable topological method, persistence barcodes, for analysing the loss landscapes of neural networks.

R&D Software Engineer, Internship

Summer 2018

Amgen Scholars Program | Zurich

- Used generative adversarial networks to characterise structural brain changes of Alzheimer's disease in MRI scans.
- Presented the method and findings at the Cambridge Amgen Scholars Symposium.

EDUCATION

MSc Computational Science and Engineering

2019 - 2022

ETH Zurich | Zurich

- Focus on robotics. Master's thesis in the Computer Vision Lab, graded 5.75 out of 6.
- Supported by the ETH Zurich Master's Scholarship, covering full tuition and living expenses.

BSc Computer Science and Electrical Engineering

2015 - 2019

Moscow Institute of Physics and Technology | Moscow

- Focus on data science. GPA 9.0 out of 10, graduated with distinction.
- Lecturer at the MIPT Education Center, 2016 - 2017: designed and taught Olympiad courses in electrostatics and geometry, and wrote a problem book of original questions with solutions.

Spoken languages: English (C1), German (B1), Russian (native)

SELECTED PROJECTS

Pixel Morph | 2026

Two neural networks that generate drawings and 3D objects live in the browser on WebGPU. No server, no upload. Describe an object in words and a new one is sampled in under half a second.

Astro Pilot | 2026

A spacecraft in Earth orbit learns to fly through asteroids, comets and city skylines with reinforcement learning. It trains and renders in the browser, and two small on-device models watch the flight and explain it.

FX Regime Radar | 2026

A daily pipeline that names the market regime for EUR/USD, USD/CHF and GBP/USD, forecasts the risk that it changes and flags unusual days. It never predicts price direction, by design.

De Novo Drug Design from Protein Pockets | 2021

Semester thesis in Prof. Gisbert Schneider's Molecular Design Lab at ETH Zurich. Euclidean neural networks encode a 3D binding pocket and a language model writes new candidate molecules.

SELECTED PUBLICATIONS AND PATENT APPLICATION

A method for determining the 6D pose of a camera used to acquire an image of a scene using a point cloud of the scene and features

W. Abbeloos, D. Emtsev, D. P. Paudel, V. Patil, A. Obukhov, L. Van Gool
International patent application WO2023186262A1 (2023).

Dynamic Plane Convolutional Occupancy Networks

S. Lionar*, D. Emtsev*, D. Svilarkovic*, S. Peng
Winter Conference on Applications of Computer Vision (WACV) (2021).

Barcodes as Summary of Loss Function Topology

S. Barannikov, A. Korotin, D. Oganessian, D. Emtsev, E. Burnaev
Doklady Rossiiskoi Akademii Nauk. Matematika, Informatika, Protsessy Upravleniya, vol. 514, no. 2 (2023).

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, C#, C, R, MATLAB, Java

Machine learning: PyTorch, TensorFlow, TensorFlow.js, OpenCV, scikit-learn, NumPy, Pandas

3D and numerics: Eigen, Libigl, Boost, Unity

Platform: Azure ML, Azure DevOps, DVC, Docker, Google Cloud, Git, Bash

AWARDS AND HONOURS

ETH Zurich Master's Scholarship - Full tuition and living expenses (2019)

Singapore International Pre-Graduate Award - Bioinformatics (2019)

Abramov Scholarship for Academic Excellence - MIPT (2016 - 2019)

All-Russian Math and Physics Olympiad - Winner (2015)

Tournament of Towns Math Competition - Winner (2013)

International Math and Physics Olympiad, Belgrade - Silver medal (2013)